DSP · L10

Chebyshev and FIR design

Cheat sheet · Day 20 · Thu 12 Nov · pack D12

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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CHEBYSHEV (I and II, same eps and N)          [L10 s24, s27]
  dp = 10^(-Rp/20), ds = 10^(-As/20)
  eps = sqrt(1/dp^2 - 1)          (1 dB -> 0.5088, 0.5 dB -> 0.3493)
  N = acosh( sqrt(1-ds^2)/(eps*ds) ) / acosh(Ws/Wp)   ROUND UP
      acosh x = ln(x + sqrt(x^2-1)); slides print "cos^-1"
  Digital: Wi = 2 tan(wi/2) (T=1) BEFORE taking Ws/Wp
  Type I : equiripple passband, |H(0)|=1 (N odd) or dp (N even), all-pole
  Type II: equiripple stopband, zeros at W = Ws/cos((2k+1)pi/2N)
CHEB I POLES  beta = ((sqrt(1+eps^2)+1)/eps)^(1/N)
  r1 = Wp(beta^2+1)/(2beta) [jW axis], r2 = Wp(beta^2-1)/(2beta) [sigma]
  phi_k = 90 + (2k+1)180/(2N) deg ; p_k = r2 cos phi_k + j r1 sin phi_k
  K = prod(-p_k) (N odd), /sqrt(1+eps^2) (N even)
FAMILIES  order: Elliptic < Cheb I = Cheb II < Butter < Bessel
          phase: Bessel best ... Elliptic worst
LINEAR PHASE FIR  alpha=(M-1)/2, group delay = alpha
  sym: H = e^{-j alpha w} A(w)   antisym: H = e^{-j(alpha w - pi/2)} A(w)
  I sym/odd: any | II sym/even: zero z=-1 (no HP)
  III anti/odd: h(alpha)=0, zeros z=+-1 (BP only) | IV anti/even: zero z=1 (no LP)
  Fill: compare ends -> sym or anti; mirror about alpha
WINDOW METHOD  hd(n) = sin(wc n)/(pi n), hd(0)=wc/pi
  course sinc x = sin x / x  -> numpy: (wc/pi)*np.sinc(wc*n/pi)
  h(n) = hd(n - alpha) w(n),  n = 0..M-1
  Rect 1 | Bartlett 1-2|n-alpha|/(M-1) | Hanning 0.5(1-cos(2pi n/(M-1)))
  Hamming 0.54-0.46cos(2pi n/(M-1)) | Blackman 0.42-0.5cos(.)+0.08cos(2.)
  M=5: Hanning=Bartlett={0,.5,1,.5,0}; Hamming={.08,.54,1,.54,.08}
  stopband: rect -21, Bart -26, Hanning -44, Hamm -53, Black -75 dB
  A(w) = h(alpha) + 2 sum h(n) cos((alpha-n)w)  -> check at 0, wc, pi
FREQ TRANSFORMS (analogue)  HP: s -> Wp Wp'/s
  BP: s -> Wp (s^2+Wl Wu)/(s(Wu-Wl)), centre sqrt(Wl Wu), order doubles
  Digital HP: z^-1 -> -(z^-1+a)/(1+a z^-1), a = -cos((wp+wp')/2)/cos((wp-wp')/2)
  Check: new edge has the same gain as the prototype edge
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